

RS003 Week 1 — English Worksheet

Topic: Open a Pygame Window · **Course:** Pygame Space Shooter · **Time:** about 45 minutes

This week you think about your very first **Pygame window** — a blank screen your game will live in. On this worksheet you read code, predict it, fix it, and explain it in your own words. The screen is a space measured in **width** and **height**, filled with a deep-space colour, and kept open with a **game loop**.

Words to know: **import**, **window**, **screen**, **fill**, **flip**, **game loop**

1 · Predict 🧠

Read each piece of code. Do **not** run it. Just from reading, write what you think will happen.

```
screen = pygame.display.set_mode((800, 600))
```

What does **(800, 600)** mean? Which number is across, which is down?


```
SPACE = (10, 12, 40)
screen.fill(SPACE)
```

The screen gets filled with a colour. But will you actually SEE it yet? What is still missing?


```
running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False
```

What is this loop watching for? What makes the window finally close?


```
pygame.display.set_caption("Space Shooter")
```

Where do you think these words will appear on the screen?

2 · Spot the Bug 🐛

Each block below was meant to do something but is broken. Read what it is **supposed** to do, then fix it on paper and explain why the original was wrong.

Bug A — This should open a window, but nothing appears. The screen is filled, but it never shows up.

```
import pygame
pygame.init()
screen = pygame.display.set_mode((800, 600))
screen.fill((10, 12, 40))
```

Hint: filling the screen is not enough. One line makes it visible.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — This is supposed to make an **800 wide, 600 tall** window. Right now it is tall and thin instead of wide.

```
screen = pygame.display.set_mode((600, 800))
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The window opens but freezes and you cannot close it with the X button.

```
import pygame
pygame.init()
screen = pygame.display.set_mode((800, 600))
screen.fill((10, 12, 40))
pygame.display.flip()

while True:
    pass
```

Hint: the loop never listens for the QUIT event.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Explain the Code

Read this working program carefully. Do **not** run it — answer every question just by reading it line by line.

```
import pygame

pygame.init()
screen = pygame.display.set_mode((800, 600))
pygame.display.set_caption("Space Shooter")

SPACE = (10, 12, 40)

running = True
while running:
    for event in pygame.event.get():
        if event.type == pygame.QUIT:
            running = False
    screen.fill(SPACE)
    pygame.display.flip()

pygame.quit()
```

What does the line `import pygame` do, and why must it come first?

What does `set_mode((800, 600))` create, and what would change if you swapped the two numbers to `(600, 800)` ?

Which line actually makes the new picture show up on the window?

Trace the variable `running` . What value does it start as, and which line changes it?

The `screen.fill(SPACE)` line is inside the loop, not above it. Why do you think it sits inside the loop?

4 · Explain Your Lesson Code 🎥

In today's live lesson you wrote your own Pygame window code. Now explain **your own code** — the one you made in class. Record a short video on your phone (or a parent's phone) and talk through it like you are teaching someone who has never seen Pygame. You can show your window opening and closing if you like. Try to use these words: **window, screen, fill, flip, game loop**.

1. Read your `set_mode` line out loud and say what the two numbers mean.
2. Point at the line that fills your screen with colour, and say which colour you chose.
3. Show or say which line makes the colour actually appear (**flip**).
4. Say which line lets the window close when you click the X.

Write what you will say in your video. Plan it here before you record — you can read from it while filming.

Submit ✓

Send this worksheet + a video explaining your lesson code to teacher on KakaoTalk.